

$$U_o, I, I_L, I_o = ?$$

$$\frac{1V - 0}{10k} = \frac{0 - U_o}{120k}$$

$$U_o = -\frac{120k}{10k} \cdot 1V$$

$$U_o = -12V$$

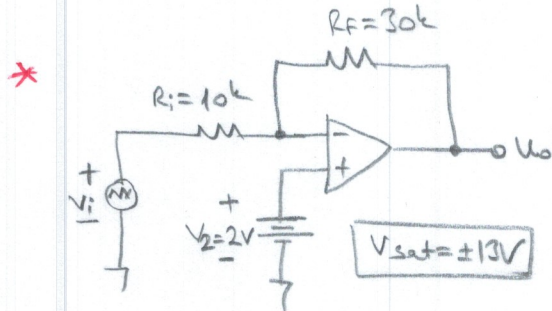
$$I = \frac{V_o - 0}{120k} \Rightarrow I = \frac{-12V}{120k}$$

$$I = -0,1mA$$

$$I_L = \frac{V_o - 0}{2k} = \frac{-12V}{2k}$$

$$I_L = -6mA$$

$$I_o = I_L + I = -6,1mA$$



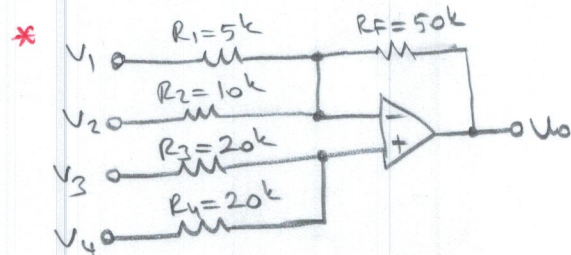
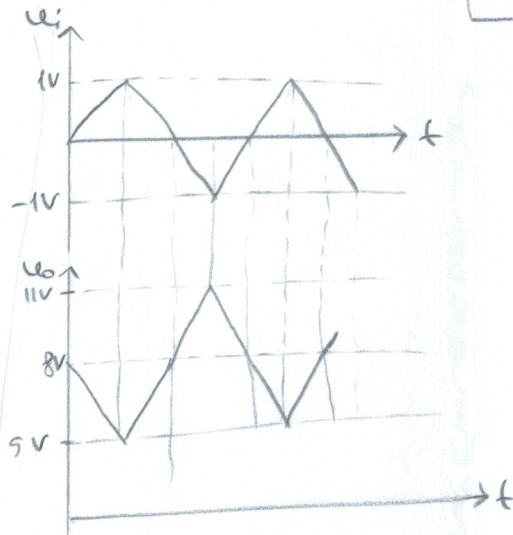
U_o 'ı elde ediniz. U_i 'le

U_o 'ı eş zamanlı olarak çiziniz.

$$U_o = -\frac{R_f}{R_i} \cdot V_i + \left(1 + \frac{R_f}{R_i}\right) \cdot V_2$$

$$U_o = -3V_i + 4 \cdot 2V$$

$$U_o = 8V - 3V_i$$



$$V_1 = 0,5V$$

$$V_2 = 500 \sin(6280t) \text{ mV}$$

$$V_3 = 100 \sin(6280t) \text{ mV}$$

$$V_4 = 1V$$

$$U_o = U_{o1} + U_{o2} + U_{o3} + U_{o4}$$

$$U_{o1} = -\frac{R_f}{R_1} \cdot V_1 = -0,5V \cdot \frac{50k}{5k} \Rightarrow U_{o1} = -5V$$

$$U_{o2} = -\frac{R_f}{R_2} \cdot V_2 = -0,5 \cdot \sin(6280t) V \cdot \frac{50k}{10k} \Rightarrow U_{o2} = -2,5 \sin(6280t) V$$

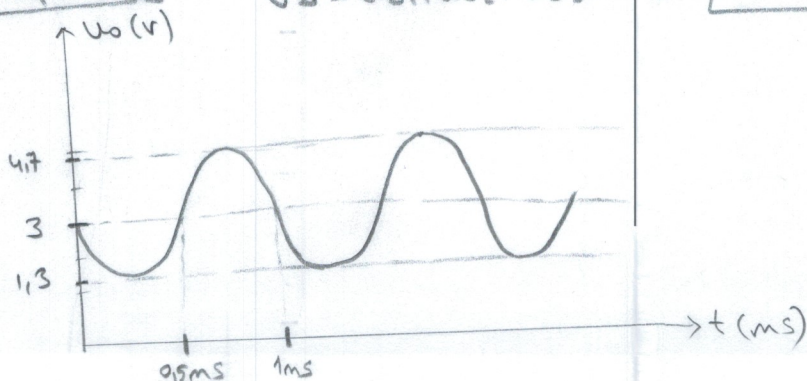
$$U_{o3} = \left(1 + \frac{R_f}{R_1 \parallel R_2}\right) \cdot \left(\frac{R_4}{R_3 + R_4}\right) \cdot V_3 = \left(1 + \frac{50k}{10k \parallel 5k}\right) \cdot \left(\frac{20k}{20k + 20k}\right) \cdot 0,1 \sin(6280t) V$$

$$U_{o3} = 0,8 \sin(6280t) V$$

$$U_{o4} = \left(1 + \frac{R_f}{R_1 \parallel R_2}\right) \cdot \left(\frac{R_3}{R_3 + R_4}\right) \cdot V_4 = \left(1 + \frac{50k}{10k \parallel 5k}\right) \cdot \left(\frac{20k}{20k + 20k}\right) \cdot 1V$$

$$U_{o4} = 8V$$

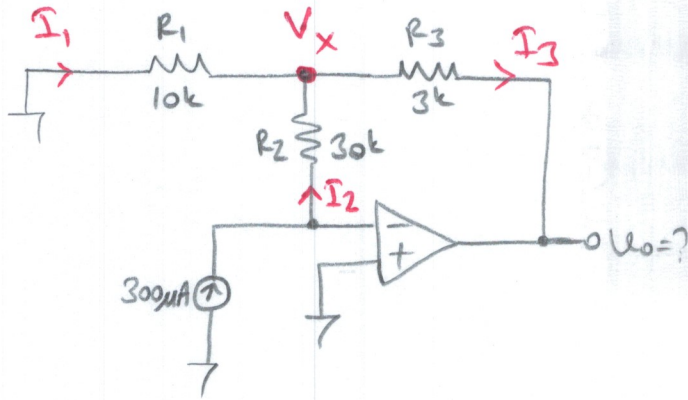
$$U_o = U_{o1} + U_{o2} + U_{o3} + U_{o4} \Rightarrow U_o = (3 - 1,7 \sin(6280t)) V$$



$$2\pi f = 6280 \Rightarrow f = 1kHz$$

$$T = 1ms$$

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$$I_2 = 300 \mu A$$

$$V_x = 0 - 300 \mu A \cdot 30k$$

$$V_x = -9V$$

$$I_1 = \frac{0 - (-9V)}{10k} = 0,9 \text{ mA}$$

$$I_3 = I_2 + I_1 = 0,3 \text{ mA} + 0,9 \text{ mA}$$

$$I_3 = 1,2 \text{ mA}$$

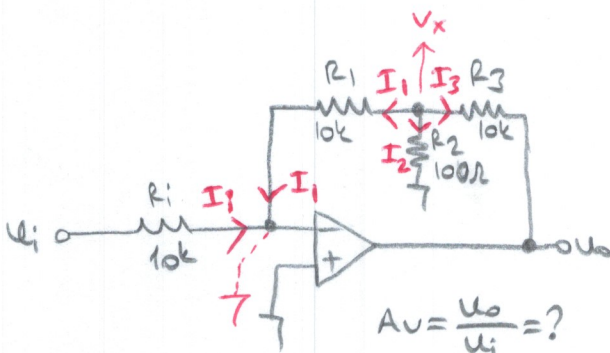
$$I_3 = \frac{V_x - V_o}{R_3}$$

$$1,2 \text{ mA} = \frac{-9V - V_o}{3k}$$

$$3,6V = -9V - V_o$$

$$V_o = -12,6V$$

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$$I_1 + I_2 + I_3 = 0$$

$$\frac{V_x - 0}{R_1} + \frac{V_x - 0}{R_2} + \frac{V_x - V_o}{R_3} = 0$$

$$V_x \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) + V_o \left(-\frac{1}{R_3} \right) = 0 \quad \text{--- (i)}$$

$$I_1 + I_i = 0$$

$$\frac{V_x - 0}{R_1} + \frac{V_i - 0}{R_i} = 0$$

$$V_x = -\frac{R_1}{R_i} \cdot V_i \quad \text{--- (ii)}$$

(ii) denklemini (i) de yerine yazarsak;

$$-\frac{R_1}{R_i} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) V_i + V_o \left(-\frac{1}{R_3} \right) = 0$$

$$\left(-\frac{1}{R_3} \right) V_o = \frac{R_1}{R_i} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) V_i$$

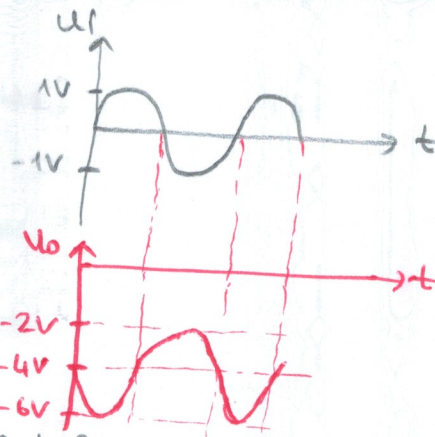
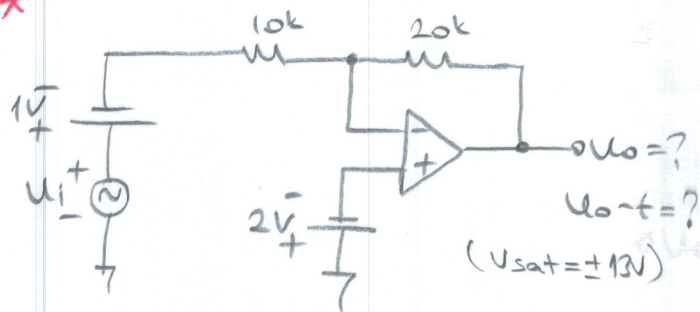
$$A_v = \frac{V_o}{V_i} = -\frac{R_1 R_3}{R_i} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

$$A_v = -\left(\frac{R_3}{R_i} + \frac{R_1 R_3}{R_i R_2} + \frac{R_1}{R_i} \right)$$

$$A_v = -\left(\frac{10k}{10k} + \frac{10k \cdot 10k}{10k \cdot 100} + \frac{10k}{10k} \right)$$

$$A_v = -102$$

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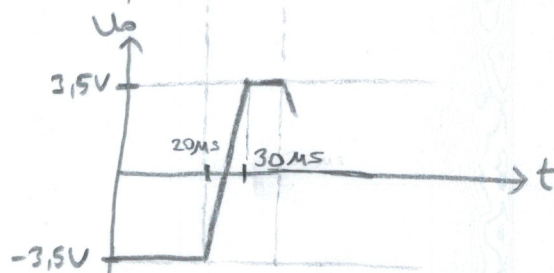
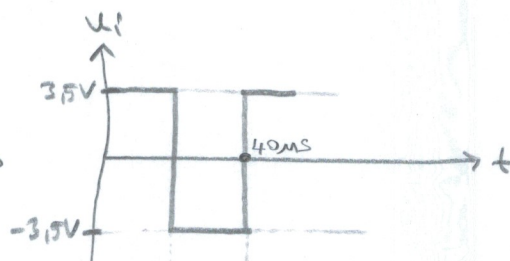
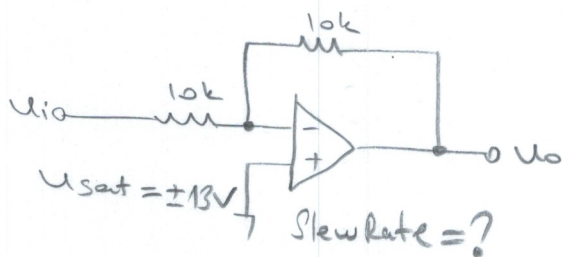


$$U_o = (U_i - 1V) \cdot \left(-\frac{20k}{10k}\right) + (-2V) \cdot \left(1 + \frac{20k}{10k}\right)$$

$$U_o = -2U_i + 2V + (-6V)$$

$$U_o = -2U_i - 4V$$

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$$SR = \frac{V_{pp}}{t_r}$$

$$SR = \frac{3.5V - (-3.5V)}{30\mu s - 20\mu s}$$

$$SR = \frac{7V}{10\mu s} \Rightarrow SR = 0.7 V/\mu s$$